Arista Access Node KOR-ACC-66 Hardware/Environmental Fault SOP

Node: KOR-ACC-66 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_9661

Description

The failure domain of hardware/environmental faults on Arista Access Nodes is primarily attributed to the occurrence of ASIC memory parity errors or line card crashes, often resulting from power supply redundancy loss. In the context of the Arista EOS, the Arista 7150 series Access Switches employ a distributed architecture, where multiple ASICs work in tandem to process traffic. The ASICs utilize a parity-based memory protection scheme to prevent data corruption. However, if a parity error occurs, the affected ASIC may crash or malfunction, leading to a loss of connectivity and network instability. Furthermore, the power supply redundancy ensures that the node remains operational even in the event of a single power supply failure. Nonetheless, if both power supplies fail or are compromised, the node may experience a loss of redundancy, exacerbating the failure domain. In the case of KOR-ACC-66, the Site Location at Koramangala Datacenter is critical, as any equipment failure can have far-reaching consequences on the network's uptime and performance.

Pre-requisite Checks

- Prerequisite: Verify redundant link status
- Prerequisite: Check for any ongoing maintenance or configuration changes
- Prerequisite: Ensure remote hands are dispatched to Koramangala Datacenter

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system resource | include Memory, Expected Output: Memory: 4096 MB (used: 1024 MB), Description: This command isolates the system's resource utilization, specifically the memory usage, which can indicate if an ASIC memory parity error has occurred.

Command: show hardware linecard | include Status, Expected Output: Status: Up, Description: This command verifies the status of the line card, which can indicate if a line card crash has occurred.

Command: show power redundancy, Expected Output: Redundancy: Yes, Description: This command checks the power supply redundancy status, which can indicate if a power supply redundancy loss has occurred.
```

Resolution Steps

Step 1: Step: Replace the affected ASIC, Command: no switch , Description: This step involves replacing the affected ASIC, which can be done by executing the 'no switch' command with the ASIC name as an argument.

Step 2: Step: Replace the failed line card, Command: no hardware linecard , Description: This step involves replacing the failed line card, which can be done by executing the 'no hardware linecard' command with the line card name as an argument.

Step 3: Step: Verify power supply redundancy, Command: show power redundancy, Description: This step involves verifying that the power supply redundancy is functioning correctly, which can be done by executing the 'show power redundancy' command.

Post-Check Verification

```
Command: show system resource | include Memory, Expected Output:
Memory: 4096 MB (used: 1024 MB), Description: This command verifies
that the system's resource utilization, specifically the memory usage,
is within normal limits.

Command: show hardware linecard | include Status, Expected Output:
Status: Up, Description: This command verifies that the line card
status is Up, indicating that the replacement was successful.

Command: show power redundancy, Expected Output: Redundancy: Yes,
Description: This command verifies that the power supply redundancy is
functioning correctly.
```

Roll-back Procedures

- Step: Restore the original ASIC, Command: switch , Description: This step involves restoring the original ASIC, which can be done by executing the 'switch' command with the ASIC name as an argument.

- Step: Restore the original line card, Command: hardware linecard , Description: This step involves restoring the original line card, which can be done by executing the 'hardware linecard' command with the line card name as an argument.

- Step: Verify power supply redundancy, Command: show power redundancy, Description: This step involves verifying that the power supply redundancy is functioning correctly.